like motion, health of muscles and nerve cells, or changes in sweat gland activity, which can reflect the intensity of a person's emotional state. But with older patients, such skin-based sensors can often cause problems, including infection.

This is where the potential of a fingernail sensor comes into play. The team realised it might be possible to derive interesting signals from how the fingernail bends throughout the course of a day, as humans use fingers to interact with their environment, and tap into the power of artificial intelligence (AI) and machine learning to analyse and derive valuable insights from that data.

IBM's system consists of strain gauges attached to the fingernail and a small computer that samples strain values, collects accelerometer data and communicates with a smart watch. The watch also runs machine learning models to rate bradykinesia, tremor and dyskinesia, which are all symptoms of Parkinson's disease.

A wireless, battery-free, biodegradable blood flow sensor

A sensor developed by Stanford University researchers could make it easier for doctors to monitor the success of blood vessel surgery. The device monitors the flow of blood through an artery. It is biodegradable, battery-free and wireless, so it is compact and does not need to be removed. It can warn a patient's doctor if there is a blockage.

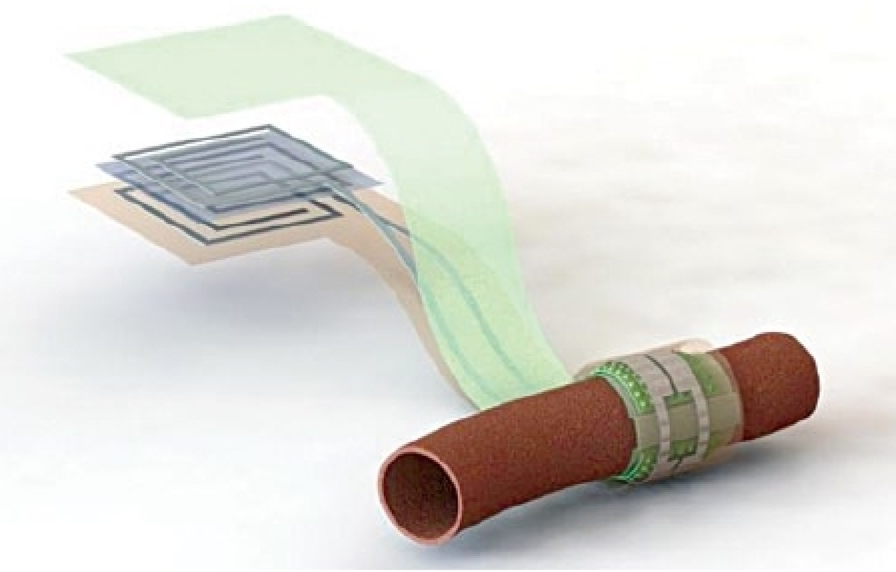

Illustration showing the biodegradable pressure sensor wrapped around a blood vessel with the antenna off to the side (Credit: Stanford University)

The sensor wraps snugly around the healing vessel, where blood pulsing past pushes on its inner surface. As the shape of that surface changes, it alters the sensor's capacity to store electric charge, which doctors can detect remotely from a device located near the skin but outside the body. That device solicits a reading by pinging the antenna of the sensor, which is similar to an ID card scanner.

The sensor was first tested in an artificial setting where the researchers pumped air through an artery-sized tube to mimic pulsing blood flow. Then, it was implanted around an artery in a rat. Even at such a small scale, the sensor successfully reported blood flow to the wireless reader.

In the future, this device could come in the form of a stick-on patch or be integrated into another technology, like a wearable device or smartphone.

Researcher uses music to manage data networks

Orchestrating traffic is a crucial component of operating data networks. Modern networks may interconnect thousands of servers, storage units or switches that, in turn, run tasks like device booting and configuration, anomaly and intrusion detection, monitoring and diagnostics. These tasks must be managed to keep the networks operating smoothly.

As networks become increasingly complex, a Saint Louis University (SLU) researcher turned to sound as a simpler alternative to manage complicated network tasks. Flavio Esposito, PhD, assistant professor of computer science at SLU, together with collaborator, Mary Hogan, a former SLU undergraduate now pursuing her doctoral degree at Princeton University, recently proposed an innovative traffic-management solution.

Esposito was interested in exploring whether a simpler network management approach could solve common problems. Ideally, Esposito says, an out-of-band management network—a type of network management that is separate from data that flows across the network—should be reliable, able to reach all devices in a data center, compatible with existing equipment, simple and inexpensive.

The researchers' answer to this is music-defined networking. Music-defined networking is a model in which network functions can be programmed in response to specific sound sequences (music), coming from real or virtual devices.

The researchers explored both active applications, where network devices were programmed to emit a certain sound, and passive applications, where sounds produced by devices, such as data centre fans, were monitored to identify when these may have failed.

Using low-cost speakers, microphones and Raspberry Pis, the duo augmented existing network components with sound capabilities. In both real and virtual network test environments, they explored how music could be used for several network tasks, including data centre server fan failure detection, authentication, load balancing and congestion notification.

Drones to tackle climate change

A team of scientists at University of Nottingham is using drones to survey woody climbing plants and better understand how these may affect the carbon balance of tropical rainforests.

Previous research has found that woody climbing plants—called lianas—have increased in both number and bulk in recent years, and are dramatically reducing the carbon uptake